

DS	Session	Linked List	Question Information	Level 1	Challenge 36
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Question description

A long time ago, there was a desolate village in India. The ancient buildings, streets, and businesses were deserted. The windows were open, and the stairwell was in disarray. You can be sure that it will be a fantastic area for mice to romp about in! People in the community have now chosen to provide high-quality education to young people in order to further the village's growth.

As a result, they established a programming language coaching centre. People from the coaching centre are presently performing a test. Create a programme for the GetNth() function, which accepts a linked list and an integer index and returns the data value contained in the node at that index position.

Example

Input: 1->10->30->14, index = 2

Output: 30

The node at index 2 is 30

Constraints

1 < N < 1000

1 < X < 1000

1 < I < 1000

Input Format

First line contains the number of datas- N.

Second line contains N integers(the given linked list).

Third Line Index I

Output Format

First Line indicates the linked list

second line indicates the node at indexing position

Logical Test Cases

Test Case 1

INPUT (STDIN)

4
1 2 3 4
2

EXPECTED OUTPUT

Linked list:-->4-->3-->2-->1
Node at index=2:3

Test Case 2

INPUT (STDIN)

4
11 21 31 41
3

EXPECTED OUTPUT

Linked list:-->41-->31-->21-->11
Node at index=3:21

Mandatory Test Cases

Test Case 1

KEYWORD

struct node

Test Case 2

KEYWORD

struct node *next;

Test Case 3

KEYWORD

int GetNth(struct node* head,int index)

Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

CYCLOMATIC COMPLEXITY

10

TOKEN COUNT

385

NLOC

74

LinkedList6.c +

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node {
5     int data;
6     struct node *next;
7 };
8
9 struct node* createNode(int data) {
10     struct node* new_node = (struct node*)malloc(sizeof(struct node));
11     new_node->data = data;
12     new_node->next = NULL;
13     return new_node;
14 }
15
16 void insertEnd(struct node** head, int data) {
17     struct node* new_node = createNode(data);
18     if (*head == NULL) {
19         *head = new_node;
20         return;
21     }
22     struct node* temp = *head;
23     while (temp->next != NULL) {
24         temp = temp->next;
25     }
```

```
26     temp->next = new_node;
27 }
28
29 int GetNth(struct node* head, int index) {
30     struct node* current = head;
31     int count = 0;
32     while (current != NULL) {
33         if (count == index) {
34             return current->data;
35         }
36         count++;
37         current = current->next;
38     }
39     return -1;
40 }
```

41

42 void reverseList(struct node** head) {

main | CH.SC.U4CSE25225 | LinkedList6.c

Ln 83, Col 1 Spaces: 4 UTF-8 CRLF C

43 struct node* prev = NULL;

44 struct node* current = *head;

45 struct node* next = NULL;

46 while (current != NULL) {

47 next = current->next;

48 current->next = prev;

49 prev = current;

50 current = next;

51 }

52 *head = prev;

53 }

54

55 void printList(struct node* head) {

```
56     struct node* p = head;
57     while (p != NULL) {
58         printf("%d%c", p->data, (p->next == NULL) ? '\n' : ' ');
59         p = p->next;
60     }
61 }
62
63 int main() {
64     int n;
65     if (scanf("%d", &n) != 1) return 0;
66
67     struct node* head = NULL;
68     for (int i = 0; i < n; i++) {
69         int val;
```

```
70     scanf("%d", &val);
71     insertEnd(&head, val);
72 }
73
74 int index;
75 scanf("%d", &index);
76 printf("Node at index=%d:%d\n", index, GetNth(head, index));
77
78 reverseList(&head);
79 printf("Reversed:");
80 printList(head);
81 return 0;
82}
83
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
4
```

```
1 2 3 4
```

```
2
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList6.exe
```

```
Node at index=2:3
```

```
Reversed:4 3 2 1
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
4
```

```
11 21 31 41
```

```
3
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList6.exe
```

```
Node at index=3:41
```

```
Reversed:41 31 21 11
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```